9.1 Sequences

A sequence is a list of numbers in a given order.

- Finite sequence: $\{a_n\}_{n=1}^{4} = \{1, 4, 7, 10\}$
- Infinite sequence: $\{a_n\}_{n=1}^{\infty} = \{3, 7, 11, 15, \ldots\}$

One can think of a sequence as a function restricted to positive integers.

In our finite sequence above, we can say we have a sequence $\{a_n\}_{n=1}^{4}$ with $a_n = 1 + 3(n-1)$; and in the case of the infinite sequence above, we have a $\{a_n\}_{n=1}^{\infty}$ sequence with $a_n = 3 + 4(n-1)$.

Representing Sequences

Definition: A **sequence** is a function a from the set of natural numbers to the set of real numbers. Instead of $a(n)$, we will usually denote the nth element in the sequence by a_n. We will use the notation $\{a_n\}$, or more precisely,

$$\{a_n\}_{n=1}^{\infty}$$

to denote a sequence.

As an example, observe that the sequence

$$\{a_n\} = \left\{1, -\frac{1}{2}, \frac{1}{3}, -\frac{1}{4}, \frac{1}{5} \cdots\right\}$$

can be written as

$$\{a_n\} = \left\{\frac{(-1)^{n+1}}{n}\right\}.$$

Convergence and Divergence

Definitions The sequence $\{a_n\}$ **converges** to the number L if for every positive number ϵ there corresponds an integer N such that for all n,

$$n > N \quad \Rightarrow \quad |a_n - L| < \epsilon.$$

If no such number L exists, we say that $\{a_n\}$ **diverges**.

The number L is said to be the limit of the sequence $\{a_n\}$, and we write $\lim_{n\to\infty} a_n = L$, or simply $a_n \to L$.

A sequence that converges is said to be **convergent**. Otherwise, the sequence is said to be **divergent**.

Example: The constant sequence $1, 1, 1, \ldots$ is convergent and the limit is 1.

Example: The sequence $\{1/n\}$ converges to 0. That is, $\lim_{n\to\infty} \frac{1}{n} = 0$.

Example:

The sequence $\{1, -1, 1, -1, \ldots, (-1)^{n+1}, \ldots\}$ is divergent.

Example: The sequence $\{\frac{n^2+1}{n^2+n}\}$ converges and

$$\lim_{n\to\infty} \frac{n^2+1}{n^2+n} = 1.$$

In some cases, a sequence diverges because it gets larger and larger. In particular, a_n will eventually become larger than any fixed number as n gets large. We use the following terms for such scenarios.

Definition The sequence $\{a_n\}$ **diverges to infinity** if for every number M there is an integer N such that for all$n$ larger that N, we have $a_n > M$. If this condition holes, we write

$$\lim_{n\to\infty} a_n = \infty \text{ or } a_n \to \infty.$$

Similarly, if for every number m there is an integer N such that for all n larger that N, we have $a_n < m$, then we say $\{a_n\}$ diverges to negative infinity and write

$$\lim_{n\to\infty} a_n = -\infty \text{ or } a_n \to -\infty.$$

Calculating Limits of Sequences

Theorem: Let $\{a_n\}$ and $\{b_n\}$ be be two convergent sequences of real numbers, and let A and B be the limits of $\{a_n\}$ and $\{b_n\}$respectively. Then the following rules hold.

1. *Sum Rule*: $\lim_{n\to\infty} (a_n + b_n) = A + B$
2. *Difference Rule*: $\lim_{n\to\infty} (a_n - b_n) = A - B$
3. *Constant Multiple Rule*: $\lim_{n\to\infty} (k \cdot b_n) = k \cdot B$ (any k)
4. *Product Rule*: $\lim_{n\to\infty} (a_n \cdot b_n) = A \cdot B$
5. *Quotient Rule*: $\lim_{n\to\infty} \left(\frac{a_n}{b_n}\right) = \frac{A}{B}$ if $B \neq 0$

Examples:

a. $\lim_{n\to\infty} (1 - \frac{1}{n}) =$

b. $\lim_{n\to\infty} (\frac{5}{n^2}) =$

c. $\lim_{n\to\infty} (\frac{4-7n^6}{n^6+3}) =$

Remark:

1. One consequence of the above theorem is that if both the sequences $\{a_n\}$ and $\{b_n\}$ converge, their sum $\{a_n + b_n\}$ also converges. However, the converse is not true. For example,

2. Another consequence is that every nonzero multiple of a divergent sequence $\{a_n\}$ is also a divergent sequence. Suppose that, to the the contrary, $\{ca_n\}$ converges for some $c \neq 0$. Then,

Theorem (The Sandwich Theorem for Sequences):

Let $\{a_n\}$, $\{b_n\}$, and $\{c_n\}$ be sequences such that $a_n \le b_n \le c_n$ for all n beyond some index N. Suppose that $\lim_{n\to\infty} a_n = \lim_{n\to\infty} c_n = L$, then $\lim_{n\to\infty} b_n = L$ also.

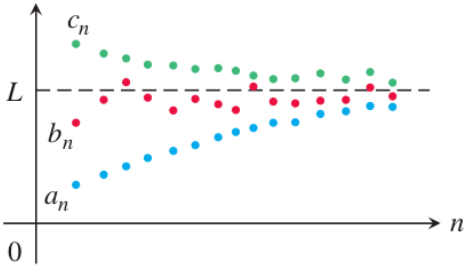


FIGURE 9.4 The terms of sequence $\{b_n\}$ are sandwiched between those of $\{a_n\}$ and $\{c_n\}$, forcing them to the same common limit L.

Example:

a. $\frac{\cos n}{n} \to 0$

b. $\frac{1}{2^n} \to 0$

c. $\frac{1}{n\sqrt{n}} \to 0$

Theorem (The Continuous Function Theorem for Sequences): Let $\{a_n\}$be a sequence of real numbers. If $a_n \to L$ and if f is a function that is continuous at L and defined at all a_n, then $f(a_n) \to f(L)$.

Example: Show that $\sqrt{(n+1)/n} \to 1$.

Example: The sequence $\{2^{1/n}\}$ converges to 1.

Using L'Hôpital's Rule

Theorem: Suppose that $f(x)$ is a function defined for all $x \geq n_0$ and that $\{a_n\}$ is a sequence of real numbers such that $a_n = f(n)$ for $n \geq n_0$. Then

$$\lim_{x\to\infty} f(x) = L \quad \Rightarrow \quad \lim_{n\to\infty} a_n = L.$$

Example: $\lim_{n\to\infty} \frac{\ln n}{n} = 0$

Example: Let $\{a_n\}$ be the sequence defined by $a_n = (\frac{n+1}{n-1})^n$. Does the sequence converge?

Commonly Occurring Limits:

1. $\lim_{n\to\infty} \sqrt[n]{n} = 1$

2. $\lim_{n\to\infty} k^{1/n} = 1$ for any fixed $k > 0$.

3. $\lim_{n\to\infty} k^{n} = 0$ for any fixed k such that $|k| < 1$.

4. $\lim_{n\to\infty} (1 + \frac{k}{n})^{n} = e^{k}$ for any fixed k.

5. $\lim_{n\to\infty} \frac{k^{n}}{n!} = 0$ for any fixed k.

Bounded Monotonic Sequences (brief)

Definition:

- A sequence $\{a_n\}$ is **bounded from above** if there exists a number M such that $a_n \leq M$ for all n.
- A sequence $\{a_n\}$ is **bounded from below** if there exists a number m such that $a_n \geq m$ for all n.
- If $\{a_n\}$ is bounded from above and below, then $\{a_n\}$ is **bounded**. Otherwise, we say $\{a_n\}$ is an **unbounded** sequence.

Definition: A sequence $\{a_n\}$ is **increasing** is $a_n \leq a_{n+1}$ for all n. The sequence $\{a_n\}$ is **decreasing** if $a_n \geq a_{n+1}$ for all n. We say a sequence $\{a_n\}$ is **monotonic** if it is either increasing or decreasing.

Theorem (The Monotonic Sequence Theorem): If a sequence $\{a_n\}$ is bounded and monotonic, the sequence converges.

Extra Examples:

1. $a_n = \frac{1-2n}{1+2n}$

2. $a_n = \frac{3^n 6^n}{2^{-n} n!}$

3. $a_n = \frac{(\ln n)^3}{n}$

4. $a_n = \sqrt{n}\sin(\frac{1}{\sqrt{n}})$